

MBDP metagenomics course

2026

Course information

- Practicals on Mon | Wed | Fri 9.00–16.00
- Self-learning materials for the days in between
- Instructions and materials can be found from:

https://github.com/MBDP-bioinformatics-courses/MBDP_Metagenomics_2024

Learning outcomes

By completing this course, you will:

- Have a basic understanding of metagenomic sequencing technologies and bioinformatic approaches to analyse metagenomic data
- Be able to plan and execute a metagenomic sequencing project depending on the research questions.
- Have an up-to-date knowledge on the bioinformatic tools and best practices for the analysis of metagenomes.
- Be able to choose and critically evaluate new tools and approaches for specific research question
- Have confidence to learn and implement new bioinformatic methods using available documentation

Introductions

- Introduce yourself briefly (<1 min)
 - Name and affiliation
 - What is your PhD about?
 - OPTIONAL: Expectations for the course?

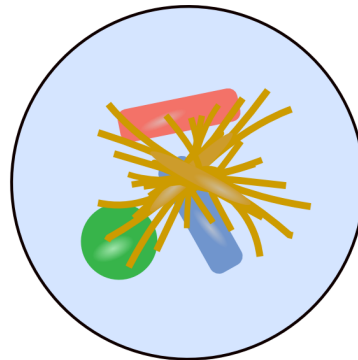
Metagenomics

Short introduction

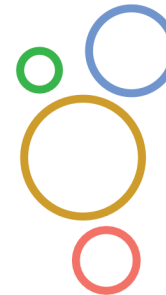
Metagenomics

- Study of all genetic material in an environment

Mixed microbial community



DNA
Extraction



Amplicon sequencing



Multiple copies of fragments
from 1 target gene

Metagenomics sequencing



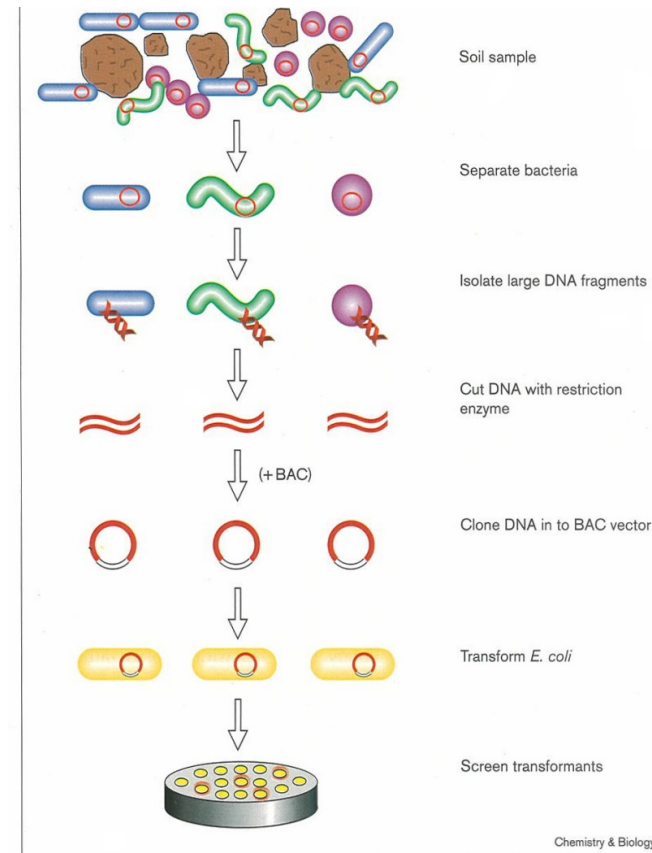
Short sequence
fragments from "all" DNA

Metagenomics

- Jo Handelsman et al. 1998

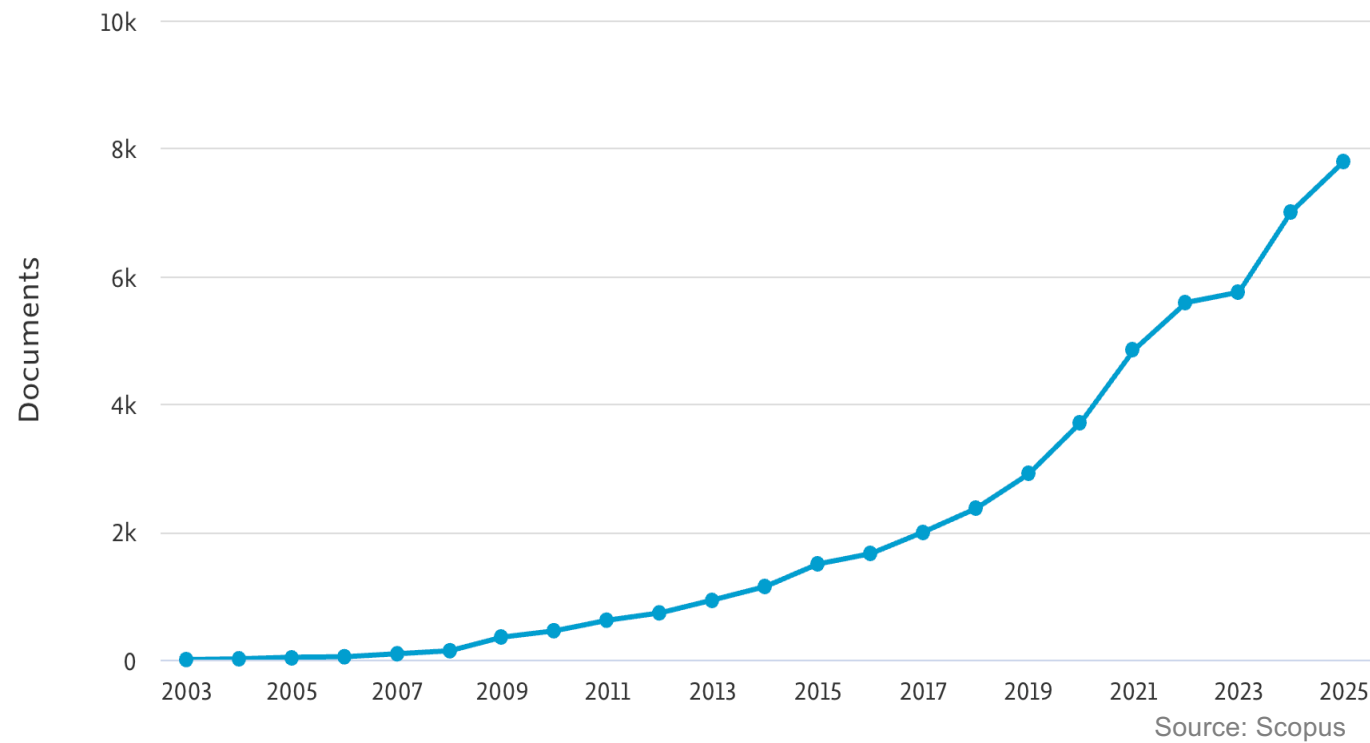
Molecular biological access to the chemistry of unknown soil microbes: a new frontier for natural products

Jo Handelsman¹, Michelle R Rondon¹, Sean F Brady², Jon Clardy² and Robert M Goodman¹



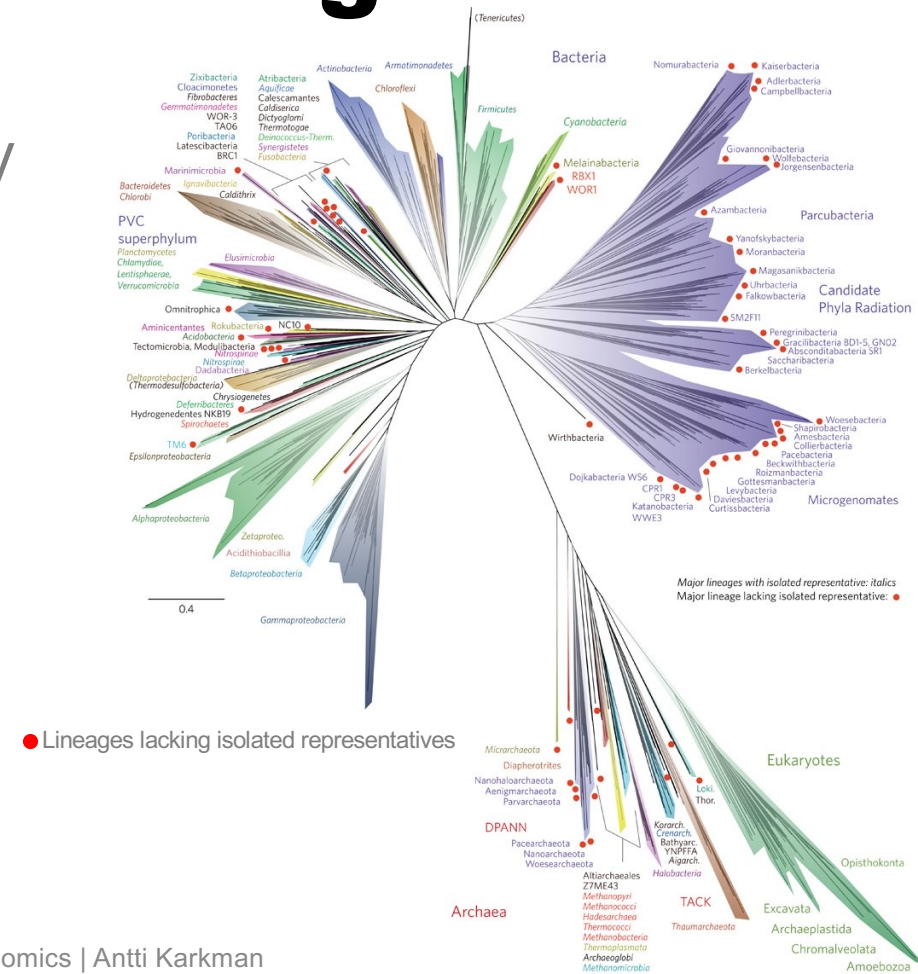
Articles published in metagenomics

Documents by year



Why do we need metagenomics?

- The great plate count anomaly
- Taxonomy $\neq$ function



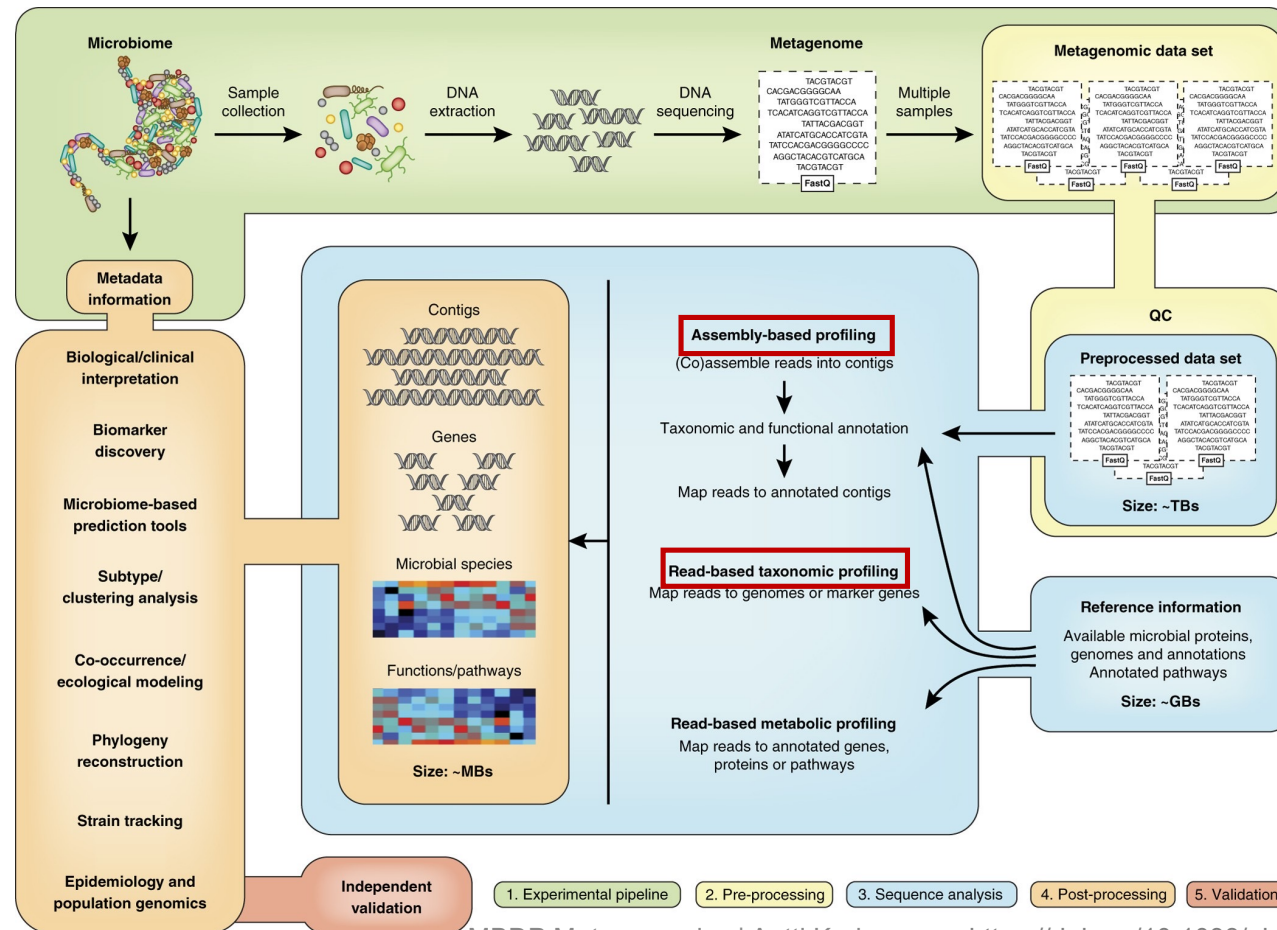
Metagenomic sequencing



Group work:

- 1-3 specific features for each technology
- Which one(s) is/are suitable for metagenomics

Metagenomic data analysis



Read-based vs. Assembly-based



Read-based

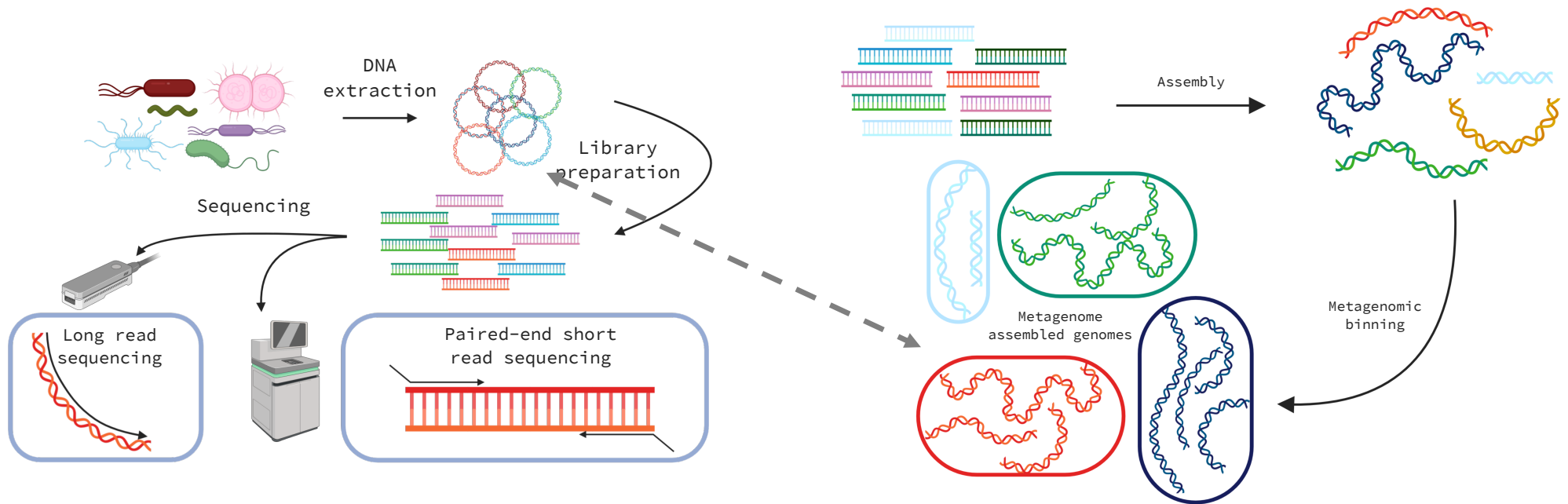


Assembly-based



Genome-resolved metagenomics

Reconstructing genomes from metagenomic sequencing data



Databases for (meta)genomics

Sequence Read Archive (SRA)

European Nucleotide Archive (ENA)

- Publicly available repositories for high-throughput sequencing data
- All sequencing data should be deposited to repositories upon publication
- Sequencing data (Runs) organised under projects (BioProject) and linked to samples (BioSample)
- Various levels of metadata depending on the project
- Web and (several) command line access options

<https://www.ncbi.nlm.nih.gov/sra> <https://www.ebi.ac.uk/ena>

MGnify

- Website to browse, analyse, discover and compare microbiome data
- Data from ENA/SRA
- Includes analyses, assemblies and MAG collections

<https://www.ebi.ac.uk/metagenomics>

Branchwater & Sandpiper

- Annotation of sequencing experiments in SRA/ENA
- Web interface to search for sequences/taxonomy

<https://sandpiper.qut.edu.au/> <https://branchwater.sourmash.bio/>

Time for practicals